

2018

AP

CollegeBoard

AP Physics C: Electricity and Magnetism

Free-Response Questions



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ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1 \text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^8$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23} \text{ mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11} (\text{N}\cdot\text{m}^2)/\text{kg}^2$
Universal gas constant, $R = 8.31 \text{ J}/(\text{mol}\cdot\text{K})$	Acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$	
1 unified atomic mass unit, $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$	
Planck's constant, $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s} = 4.14 \times 10^{-15} \text{ eV}\cdot\text{s}$	
	$hc = 1.99 \times 10^{-25} \text{ J}\cdot\text{m} = 1.24 \times 10^3 \text{ eV}\cdot\text{nm}$
Vacuum permittivity, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$	
Coulomb's law constant, $k = 1/(4\pi\epsilon_0) = 9.0 \times 10^9 (\text{N}\cdot\text{m}^2)/\text{C}^2$	
Vacuum permeability, $\mu_0 = 4\pi \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
Magnetic constant, $k' = \mu_0/(4\pi) = 1 \times 10^{-7} (\text{T}\cdot\text{m})/\text{A}$	
1 atmosphere pressure, $1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$	

UNIT SYMBOLS	meter, m	mole, mol	watt, W	farad, F
	kilogram, kg	hertz, Hz	coulomb, C	tesla, T
	second, s	newton, N	volt, V	degree Celsius, °C
	ampere, A	pascal, Pa	ohm, Ω	electron volt, eV
	kelvin, K	joule, J	henry, H	

PREFIXES		
Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following assumptions are used in this exam.

- I. The frame of reference of any problem is inertial unless otherwise stated.
- II. The direction of current is the direction in which positive charges would drift.
- III. The electric potential is zero at an infinite distance from an isolated point charge.
- IV. All batteries and meters are ideal unless otherwise stated.
- V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.



ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$$\begin{aligned}
 v_x &= v_{x0} + a_x t \\
 x &= x_0 + v_{x0} t + \frac{1}{2} a_x t^2 \\
 v_x^2 &= v_{x0}^2 + 2a_x(x - x_0) \\
 \vec{a} &= \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m} \\
 \vec{F} &= \frac{d\vec{p}}{dt} \\
 \vec{J} &= \int \vec{F} dt = \Delta \vec{p} \\
 \vec{p} &= m\vec{v} \\
 |\vec{F}_f| &\leq \mu |\vec{F}_N| \\
 \Delta E &= W = \int \vec{F} \cdot d\vec{r} \\
 K &= \frac{1}{2} mv^2 \\
 P &= \frac{dE}{dt} \\
 P &= \vec{F} \cdot \vec{v} \\
 \Delta U_g &= mg\Delta h \\
 a_c &= \frac{v^2}{r} = \omega^2 r \\
 \vec{\tau} &= \vec{r} \times \vec{F} \\
 \vec{\alpha} &= \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I} \\
 I &= \int r^2 dm = \sum mr^2 \\
 x_{cm} &= \frac{\sum m_i x_i}{\sum m_i} \\
 v &= r\omega \\
 \vec{L} &= \vec{r} \times \vec{p} = I\vec{\omega} \\
 K &= \frac{1}{2} I\omega^2 \\
 \omega &= \omega_0 + \alpha t \\
 \theta &= \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2
 \end{aligned}$$

$$\begin{aligned}
 a &= \text{acceleration} \\
 E &= \text{energy} \\
 F &= \text{force} \\
 f &= \text{frequency} \\
 h &= \text{height} \\
 I &= \text{rotational inertia} \\
 J &= \text{impulse} \\
 K &= \text{kinetic energy} \\
 k &= \text{spring constant} \\
 \ell &= \text{length} \\
 L &= \text{angular momentum} \\
 m &= \text{mass} \\
 P &= \text{power} \\
 p &= \text{momentum} \\
 r &= \text{radius or distance} \\
 T &= \text{period} \\
 t &= \text{time} \\
 U &= \text{potential energy} \\
 v &= \text{velocity or speed} \\
 W &= \text{work done on a system} \\
 x &= \text{position} \\
 \mu &= \text{coefficient of friction} \\
 \theta &= \text{angle} \\
 \tau &= \text{torque} \\
 \omega &= \text{angular speed} \\
 \alpha &= \text{angular acceleration} \\
 \phi &= \text{phase angle} \\
 \vec{F}_s &= -k\Delta x \\
 U_s &= \frac{1}{2} k(\Delta x)^2 \\
 x &= x_{max} \cos(\omega t + \phi) \\
 T &= \frac{2\pi}{\omega} = \frac{1}{f} \\
 T_s &= 2\pi \sqrt{\frac{m}{k}} \\
 T_p &= 2\pi \sqrt{\frac{\ell}{g}} \\
 |\vec{F}_G| &= \frac{Gm_1 m_2}{r^2} \\
 U_G &= -\frac{Gm_1 m_2}{r}
 \end{aligned}$$

ELECTRICITY AND MAGNETISM

$$\begin{aligned}
 |\vec{F}_E| &= \frac{1}{4\pi\epsilon_0} \left| \frac{q_1 q_2}{r^2} \right| \\
 \vec{E} &= \frac{\vec{F}_E}{q} \\
 \oint \vec{E} \cdot d\vec{A} &= \frac{Q}{\epsilon_0} \\
 E_x &= -\frac{dV}{dx} \\
 \Delta V &= -\int \vec{E} \cdot d\vec{r} \\
 V &= \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i} \\
 U_E &= qV = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \\
 \Delta V &= \frac{Q}{C} \\
 C &= \frac{\kappa\epsilon_0 A}{d} \\
 C_p &= \sum_i C_i \\
 \frac{1}{C_s} &= \sum_i \frac{1}{C_i} \\
 I &= \frac{dQ}{dt} \\
 U_C &= \frac{1}{2} Q\Delta V = \frac{1}{2} C(\Delta V)^2 \\
 R &= \frac{\rho\ell}{A} \\
 \vec{E} &= \rho\vec{J} \\
 I &= Nev_d A \\
 I &= \frac{\Delta V}{R} \\
 R_s &= \sum_i R_i \\
 \frac{1}{R_p} &= \sum_i \frac{1}{R_i} \\
 P &= I\Delta V
 \end{aligned}$$

$$\begin{aligned}
 A &= \text{area} \\
 B &= \text{magnetic field} \\
 C &= \text{capacitance} \\
 d &= \text{distance} \\
 E &= \text{electric field} \\
 \mathcal{E} &= \text{emf} \\
 F &= \text{force} \\
 I &= \text{current} \\
 J &= \text{current density} \\
 L &= \text{inductance} \\
 \ell &= \text{length} \\
 n &= \text{number of loops of wire per unit length} \\
 N &= \text{number of charge carriers per unit volume} \\
 P &= \text{power} \\
 Q &= \text{charge} \\
 q &= \text{point charge} \\
 R &= \text{resistance} \\
 r &= \text{radius or distance} \\
 t &= \text{time} \\
 U &= \text{potential or stored energy} \\
 V &= \text{electric potential} \\
 v &= \text{velocity or speed} \\
 \rho &= \text{resistivity} \\
 \Phi &= \text{flux} \\
 \kappa &= \text{dielectric constant} \\
 \vec{F}_M &= q\vec{v} \times \vec{B} \\
 \oint \vec{B} \cdot d\vec{\ell} &= \mu_0 I \\
 d\vec{B} &= \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \hat{r}}{r^2} \\
 \vec{F} &= \int I d\vec{\ell} \times \vec{B} \\
 B_s &= \mu_0 n I \\
 \Phi_B &= \int \vec{B} \cdot d\vec{A} \\
 \mathcal{E} &= \oint \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt} \\
 \mathcal{E} &= -L \frac{dI}{dt} \\
 U_L &= \frac{1}{2} LI^2
 \end{aligned}$$



ADVANCED PLACEMENT PHYSICS C EQUATIONS

GEOMETRY AND TRIGONOMETRY

Rectangle

$$A = bh$$

Triangle

$$A = \frac{1}{2}bh$$

Circle

$$A = \pi r^2$$

$$C = 2\pi r$$

$$s = r\theta$$

Rectangular Solid

$$V = \ell wh$$

Cylinder

$$V = \pi r^2 \ell$$

$$S = 2\pi r \ell + 2\pi r^2$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Triangle

$$a^2 + b^2 = c^2$$

$$\sin \theta = \frac{a}{c}$$

$$\cos \theta = \frac{b}{c}$$

$$\tan \theta = \frac{a}{b}$$

A = area

C = circumference

V = volume

S = surface area

b = base

h = height

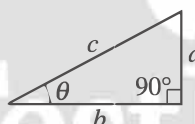
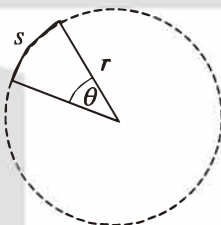
ℓ = length

w = width

r = radius

s = arc length

θ = angle



CALCULUS

$$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a \cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$$

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$$

VECTOR PRODUCTS

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$





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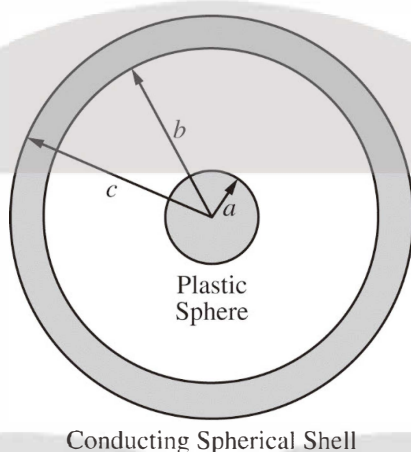
PHYSICS C: ELECTRICITY AND MAGNETISM

SECTION II

Time—45 minutes

3 Questions

Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.



1. A solid plastic sphere of radius a and a conducting spherical shell of inner radius b and outer radius c are shown in the figure above. The shell has an unknown charge. The solid plastic sphere has a charge per unit volume given by $\rho(r) = \beta r$, where β is a positive constant and r is the distance from the center of the sphere. Express your answers to parts (a), (b), and (c) in terms of β , r , a , and physical constants, as appropriate.
 - (a) Consider a Gaussian sphere of radius r concentric with the plastic sphere. Derive an expression for the charge enclosed by the Gaussian sphere for the following regions.
 - i. $r < a$
 - ii. $a < r < b$
 - (b) Use Gauss's law to derive an expression for the magnitude of the electric field in the following regions.
 - i. $r < a$
 - ii. $a < r < b$
 - (c) At any point outside of the conducting shell, it is observed that the magnitude of the electric field is zero.
 - i. Determine the charge on the inner surface of the conducting shell.
Justify your answer.
 - ii. Determine the charge on the outer surface of the conducting shell.



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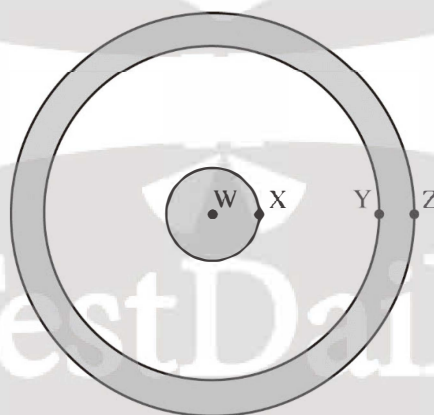
(d)

- i. On the axes below, sketch the electric field E as a function of distance r from the center of the sphere. Sketch the graph for the range $r = 0$ at the center of the sphere to $r = c$ at the outside of the conducting shell.



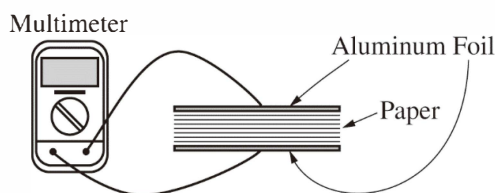
- ii. The figure below shows the sphere and shell with four points labeled W, X, Y, and Z. Point W is at the center of the sphere, point X is on the surface of the sphere, and points Y and Z are on the inner and outer surface of the shell, respectively. Rank the points according to the electric potential at that point, with 1 indicating the largest electric potential. If two points have the same electric potential, give them the same numerical ranking.

___ W ___ X ___ Y ___ Z





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2. An experiment is designed to measure the dielectric constant of paper that has an area $A = 0.060 \text{ m}^2$. Using aluminum foil, two parallel plates are created with the same area as the paper. Five hundred sheets of paper are placed between the aluminum foil plates to create a parallel plate capacitor, as shown in the figure above. Using a multimeter, the capacitance C of the capacitor is measured. The number of sheets and the total thickness d of the stack of paper are recorded. The experiment is repeated, reducing the number of sheets of paper each time. The data are recorded in the table below.

Sheets of Paper	d (m)	C (F)		
500	0.045	6.5×10^{-11}		
400	0.036	7.4×10^{-11}		
300	0.027	8.9×10^{-11}		
200	0.018	11.9×10^{-11}		
100	0.010	21.0×10^{-11}		

- (a) Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the dielectric constant of the paper.

Vertical axis: _____

Horizontal axis: _____

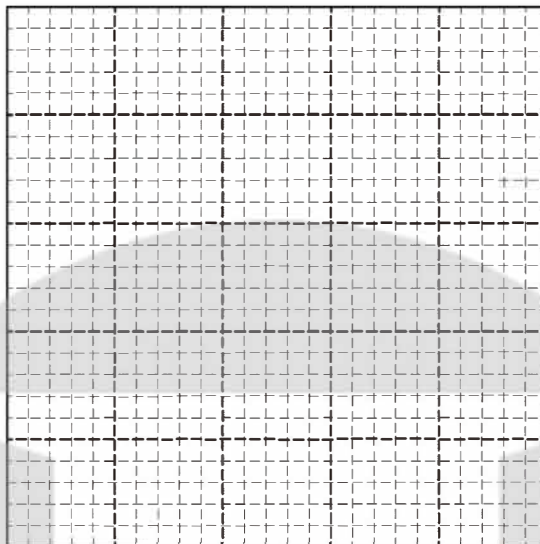
Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

TestDaily

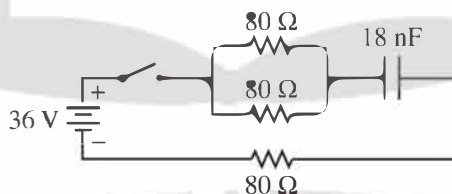


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- (b) Plot the data points for the quantities indicated in part (a) on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.



- (c) Using the straight line, calculate a dielectric constant for the paper.



The student now makes a capacitor using the same aluminum foil plates and just one sheet of paper. Using the experimentally determined dielectric constant, the student calculates the capacitance to be 18 nF. The student uses this uncharged capacitor to build a circuit using wire, a 36 V battery, 3 identical 80 Ω resistors, and an open switch, as shown in the figure above.

- (d) Calculate the current in the battery immediately after the switch is closed.
- (e) Determine the time constant for this circuit.
- (f) Students A and B measure the time it takes after the switch is closed for the voltage across the capacitor to reach half its maximum value and find that it is longer than expected.
- Student A assumes that the capacitance value is correct. Would Student A conclude that the resistance value is larger or smaller than measured?
 _____ Larger than measured _____ Smaller than measured
 Explain experimentally what could account for this.
 - Student B assumes that the resistance value is correct. Would Student B conclude that the capacitance value is larger or smaller than measured?
 _____ Larger than measured _____ Smaller than measured
 Explain experimentally what could account for this.



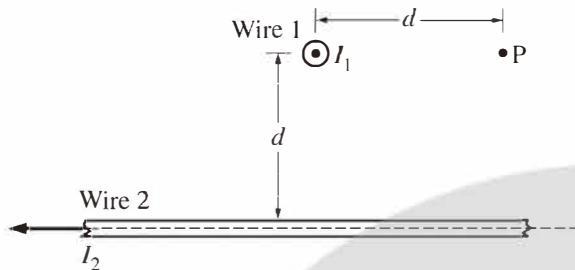
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Figure 1. Side view

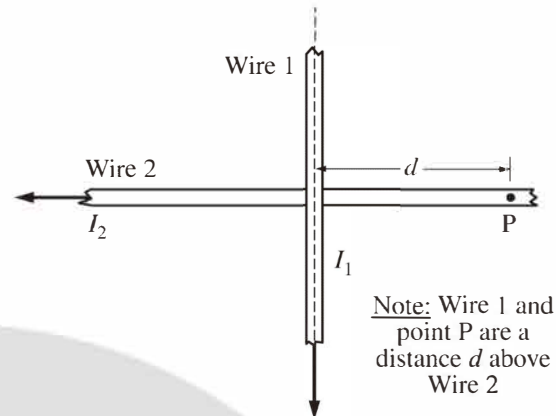


Figure 2. Top view

Note: Wire 1 and point P are a distance d above Wire 2

3. The figures above represent different views of two long, straight, horizontal wires, 1 and 2, carrying currents $I_1 = I$ and $I_2 = 2I$, respectively, in the directions shown. The wires are held in place. In Figure 1, the current in wire 1 is directed out of the page, and wire 1 is a distance d above wire 2. Point P is a horizontal distance d from wire 1 and a distance d directly above wire 2. Express your answers to parts (a) and (b) in terms of I , d , and physical constants, as appropriate.
- Use Ampere's law to derive an expression for the magnitude of the magnetic field at point P due to wire 1.
 - Derive an expression for the magnitude of the net magnetic field at point P.
 - Calculate the numerical value of the angle to the horizontal for the direction of the net magnetic field at point P.
 - Wire 1 is now released. Which of the following best describes the initial motion of wire 1 due to the magnetic field of wire 2? Assume gravitational effects are negligible.
 - ☐ Wire 1 will not move.
 - ☐ Wire 1 will move upward as viewed in Figure 1.
 - ☐ Wire 1 will move downward as viewed in Figure 1.
 - ☐ Wire 1 will rotate clockwise as viewed in Figure 2.
 - ☐ Wire 1 will rotate counterclockwise as viewed in Figure 2.

Justify your answer.



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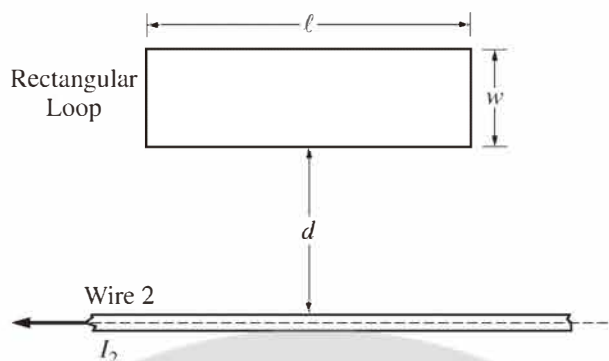


Figure 3. Side view

Wire 1 is now replaced by a conducting rectangular loop of length ℓ , width w , and resistance R . The loop is placed a distance d from wire 2, as shown. The loop, wire, and distance d are all in the plane of the page. The long side of the loop is parallel to the wire. The current I_2 for wire 2 is decreasing linearly as a function of time t according to the equation $I_2 = 2I_0(1 - kt)$, where k is a positive constant with units of s^{-1} .

(e) Of the following, select the integration that will give an expression for the flux Φ as a function of time t .

_____ $\Phi = \int_{r=d}^{r=d+w} \frac{\mu_0(2I_0)(1-kt)\ell w}{2\pi} dr$

_____ $\Phi = \int_{r=d}^{r=w} \frac{\mu_0(2I_0)(1-kt)\ell w}{2\pi} dr$

_____ $\Phi = \int_{r=d}^{r=d+w} \frac{\mu_0(2I_0)(1-kt)}{2\pi r} \ell dr$

_____ $\Phi = \int_{r=d}^{r=w} \frac{\mu_0(2I_0)(1-kt)}{2\pi r} \ell dr$

(f) Given that the flux through the rectangular loop as a function of time t is given by the equation $\Phi = \frac{\mu_0 I_0 \ell (1 - kt)}{\pi} \ln\left(\frac{d + w}{d}\right)$, derive an expression for the magnitude of the current, if any, induced in the loop. Express your answers in terms of I_0 , d , r , R , w , k , ℓ , and physical constants, as appropriate.

(g) What is the direction of the current, if any, induced in the loop as seen in Figure 3?

_____ Clockwise _____ Counterclockwise

_____ Undefined, because there is no current induced in the loop

Justify your answer.





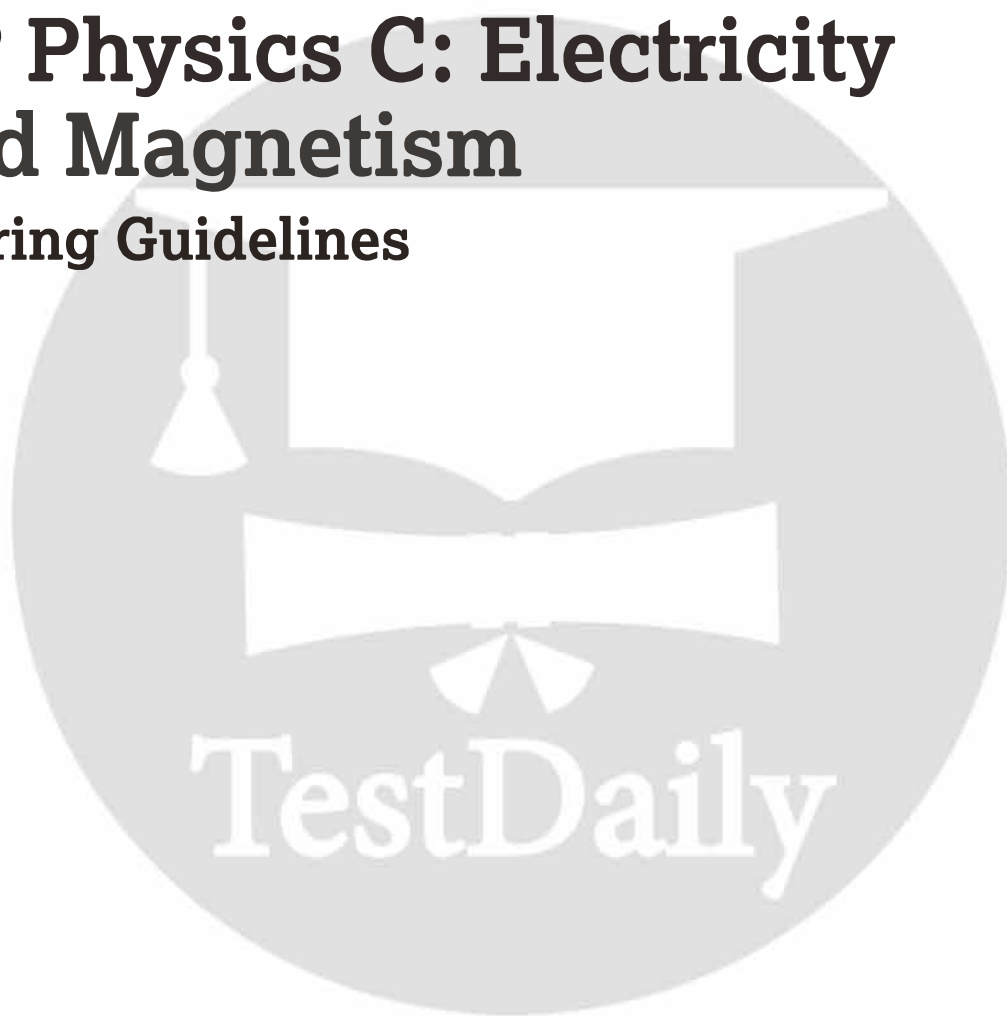
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AP Physics C: Electricity and Magnetism

Scoring Guidelines



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General Notes About 2018 AP Physics Scoring Guidelines

1. The solutions contain the most common method of solving the free-response questions and the allocation of points for this solution. Some also contain a common alternate solution. Other methods of solution also receive appropriate credit for correct work.
2. The requirements that have been established for the paragraph-length response in Physics 1 and Physics 2 can be found on AP Central at <https://secure-media.collegeboard.org/digitalServices/pdf/ap/paragraph-length-response.pdf>.
3. Generally, double penalty for errors is avoided. For example, if an incorrect answer to part (a) is correctly substituted into an otherwise correct solution to part (b), full credit will usually be awarded. One exception to this may be cases when the numerical answer to a later part should be easily recognized as wrong, e.g., a speed faster than the speed of light in vacuum.
4. Implicit statements of concepts normally receive credit. For example, if use of the equation expressing a particular concept is worth 1 point, and a student's solution embeds the application of that equation to the problem in other work, the point is still awarded. However, when students are asked to derive an expression, it is normally expected that they will begin by writing one or more fundamental equations, such as those given on the exam equation sheet. For a description of the use of such terms as “derive” and “calculate” on the exams, and what is expected for each, see “The Free-Response Sections — Student Presentation” in the *AP Physics; Physics C: Mechanics, Physics C: Electricity and Magnetism Course Description* or “Terms Defined” in the *AP Physics 1: Algebra-Based Course and Exam Description* and the *AP Physics 2: Algebra-Based Course and Exam Description*.
5. The scoring guidelines typically show numerical results using the value $g = 9.8 \text{ m/s}^2$, but the use of 10 m/s^2 is of course also acceptable. Solutions usually show numerical answers using both values when they are significantly different.
6. Strict rules regarding significant digits are usually not applied to numerical answers. However, in some cases answers containing too many digits may be penalized. In general, two to four significant digits are acceptable. Numerical answers that differ from the published answer due to differences in rounding throughout the question typically receive full credit. Exceptions to these guidelines usually occur when rounding makes a difference in obtaining a reasonable answer. For example, suppose a solution requires subtracting two numbers that should have five significant figures and that differ starting with the fourth digit (e.g., 20.295 and 20.278). Rounding to three digits will lose the accuracy required to determine the difference in the numbers, and some credit may be lost.



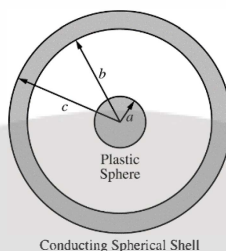


AP[®] PHYSICS C: ELECTRICITY AND MAGNETISM 2018 SCORING GUIDELINES

Question 1

15 points total

Distribution
of points



A solid plastic sphere of radius a and a conducting spherical shell of inner radius b and outer radius c are shown in the figure above. The shell has an unknown charge. The solid plastic sphere has a charge per unit volume given by $\rho(r) = \beta r$, where β is a positive constant and r is the distance from the center of the sphere. Express your answers to parts (a), (b), and (c) in terms of β , r , a , and physical constants, as appropriate.

- (a) Consider a Gaussian sphere of radius r concentric with the plastic sphere. Derive an expression for the charge enclosed by the Gaussian sphere for the following regions.

- i. 3 points

$$r < a$$

For using the integral to determine the charge enclosed	1 point
$q_{enc} = \int \rho dV = \int \beta r dV$	
For correctly substituting for dV into the integration to determine the charge enclosed	1 point
$q_{enc} = \int (4\pi r'^2)(\beta r') dr' = 4\beta\pi \int_{r'=0}^{r'=r} r'^3 dr' = 4\beta\pi \left[\frac{r'^4}{4} \right]_{r'=0}^{r'=r}$	
For a correct answer	1 point
$q_{enc} = \beta\pi r^4$	

- ii. 1 points

$$a < r < b$$

For an answer consistent with (a)(i)	1 point
$q_{enc} = \beta\pi a^4$	



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2018 SCORING GUIDELINES**

Question 1 (continued)

**Distribution
of points**

(b) Use Gauss's law to derive an expression for the magnitude of the electric field in the following regions.

i. 2 points

$$r < a$$

For correctly substituting the area of a Gaussian sphere into Gauss's law		1 point
$\frac{q_{enc}}{\epsilon_0} = \oint \vec{E} \cdot d\vec{A}$		
$\frac{q(r)}{\epsilon_0} = EA_{sphere} = E(4\pi r^2)$		
For correctly substituting the charge from part (a)(i) into the equation above		1 point
$\frac{\beta \pi r^4}{\epsilon_0} = E(4\pi r^2)$		
$E = \frac{\beta r^2}{4\epsilon_0}$		

ii. 1 point

$$a < r < b$$

For correctly substituting the charge from part (a)(ii) into Gauss's law to calculate the electric field		1 point
$\frac{\beta \pi a^4}{\epsilon_0} = E(4\pi r^2)$		
$E = \frac{\beta a^4}{4\epsilon_0 r^2}$		





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Question 1 (continued)

**Distribution
of points**

(c) At any point outside of the conducting shell, it is observed that the magnitude of the electric field is zero.

i. 2 points

Determine the charge on the inner surface of the conducting shell.
Justify your answer.

For an answer consistent with part (a)(ii) (must have opposite sign)		1 point
$q_{inner} = -\beta\pi a^4$		
For a correct justification		1 point
$E_{shell} = 0 = \frac{q_{enc}}{\epsilon_0} \therefore q_{enc} = 0 = q_{sphere} + q_{inner} \therefore q_{inner} = -q_{sphere}$		

ii. 1 point

Determine the charge on the outer surface of the conducting shell.

A correct answer		1 point
$q_{outer} = 0$		

(d)

i. 3 points

On the axes below, sketch the electric field E as a function of distance r from the center of the sphere. Sketch the graph for the range $r = 0$ at the center of the sphere to $r = c$ at the outside of the conducting shell.



For concave up graph that begins at the origin and increases to $r = a$		1 point
For a concave up graph that is continuous with the first section and decreases from $r = a$ to nonzero value at $r = b$		1 point
For a graph with a value of $E = 0$ for $r > b$		1 point
Note: Due to scale, discontinuity at point b is not required.		



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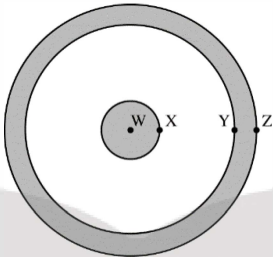
Question 1 (continued)

Distribution
of points

- (d) (continued)
ii. 2 points

The figure below shows the sphere and shell with four points labeled W, X, Y, and Z. Point W is at the center of the sphere, point X is on the surface of the sphere, and points Y and Z are on the inner and outer surface of the shell, respectively. Rank the points according to the electric potential at that point, with 1 indicating the largest electric potential. If two points have the same electric potential, give them the same numerical ranking.

__1__ W __2__ X __3__ Y __3__ Z



For any ranking that has Y and Z at the same potential	1 point
For ranking the electric potentials: $W > X > Y \text{ \& } Z$ (because $Y = Z$ is the first point)	1 point



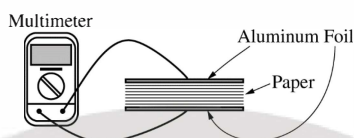


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Question 2

15 points total

Distribution
of points



An experiment is designed to measure the dielectric constant of paper that has an area $A = 0.060 \text{ m}^2$. Using aluminum foil, two parallel plates are created with the same area as the paper. Five hundred sheets of paper are placed between the aluminum foil plates to create a parallel plate capacitor, as shown in the figure above. Using a multimeter, the capacitance C of the capacitor is measured. The number of sheets and the total thickness d of the stack of paper are recorded. The experiment is repeated, reducing the number of sheets of paper each time. The data are recorded in the table below.

Sheets of Paper	d (m)	C (F)		
500	0.045	6.5×10^{-11}		
400	0.036	7.4×10^{-11}		
300	0.027	8.9×10^{-11}		
200	0.018	11.9×10^{-11}		
100	0.010	21.0×10^{-11}		

(a) 1 point

Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the dielectric constant of the paper.

Vertical axis: _____

Horizontal axis: _____

Use the remaining columns in the table above, as needed, to record any quantities that you indicated that are not given. Label each column you use and include units.

For indicating variables that will create a straight line whose slope can be used to determine the dielectric constant of the paper		1 point
Example: Vertical axis: C Horizontal axis: $\frac{1}{d}$		
Note: Student earns full credit if the axes are reversed or if they use another acceptable combination.		



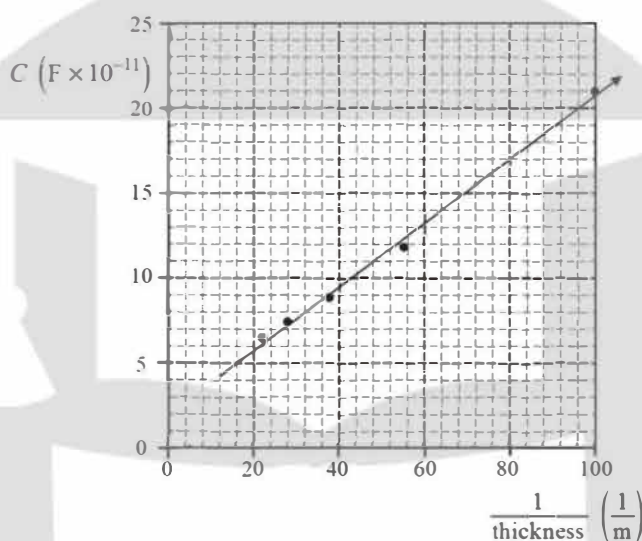
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Question 2 (continued)

Distribution
of points

(b) 4 points

Plot the data points for the quantities indicated in part (a) on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.



For a correct scale that uses more than half the grid	1 point
For correctly labeling the axes including units	1 point
For correctly plotting the data	1 point
For drawing a straight line consistent with the plotted data	1 point

(c) 3 points

Using the straight line, calculate a dielectric constant for the paper.

For correctly calculating the slope from the best-fit line and not the data points unless the points fall on the best-fit line	1 point
$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{(17 - 8)(\text{F} \times 10^{-11})}{(80 - 32)(1/\text{m})} = 0.19 \text{ F} \cdot \text{m} \quad (\text{Linear regression} = 0.187 \text{ F} \cdot \text{m})$	
For correctly relating the slope to the dielectric constant	1 point
$\text{slope} = \kappa \epsilon_0 A$	
$\kappa = \frac{\text{slope}}{\epsilon_0 A} = \frac{(0.19 \text{ F} \cdot \text{m})}{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.060 \text{ m}^2)}$	
For a correct answer	1 point
$\kappa = 3.58 \quad (\text{Linear regression} = 3.52)$	

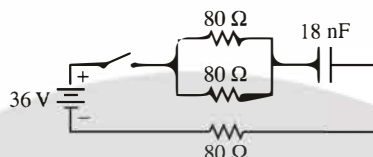




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Question 2 (continued)

**Distribution
of points**



The student now makes a capacitor using the same aluminum foil plates and just one sheet of paper. Using the experimentally determined dielectric constant, the student calculates the capacitance to be 18 nF. The student uses this uncharged capacitor to build a circuit using wire, a 36 V battery, 3 identical 80 Ω resistors, and an open switch, as shown in the figure above.

(d) 3 points

Calculate the current in the battery immediately after the switch is closed.

For calculating the equivalent resistance for the parallel resistors	1 point
$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{80\ \Omega} + \frac{1}{80\ \Omega} = \frac{2}{80\ \Omega} = \frac{1}{40\ \Omega}$	
$R_p = 40\ \Omega$	
For using Ohm's law with the potential difference across the capacitor equal to zero	1 point
$I = \frac{V}{R} = \frac{V}{R_p + R_3} = \frac{(36\ \text{V})}{R_p + R_3}$	
For substitution of values for resistance including the value for combined resistance above	1 point
$I = \frac{V}{R_p + R_3} = \frac{(36\ \text{V})}{(40\ \Omega + 80\ \Omega)} = 0.30\ \text{A}$	

(e) 2 points

Determine the time constant for this circuit.

For using the equation for the time constant with the equivalent resistance from above	1 point
$\tau = R_{eq}C = (120\ \Omega)(18\ \text{nF})$	
For an answer with units consistent with part (d)	1 point
$t = 2.16\ \mu\text{s}$	



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Question 2 (continued)

**Distribution
of points**

- (f) Students A and B measure the time it takes after the switch is closed for the voltage across the capacitor to reach half its maximum value and find that it is longer than expected.

i. 1 point

Student A assumes that the capacitance value is correct. Would Student A conclude that the resistance value is larger or smaller than measured?

____ Larger than measured ____ Smaller than measured

Explain experimentally what could account for this.

Select “Larger than measured”		
For an appropriate explanation		1 point
<i>Example: The battery is not ideal and has internal resistance, so the actual resistance for the circuit is larger than the measured resistance.</i>		

ii. 1 points

Student B assumes that the resistance value is correct. Would Student B conclude that the capacitance value is larger or smaller than measured?

____ Larger than measured ____ Smaller than measured

Explain experimentally what could account for this.

Select “Larger than measured”		
For an appropriate explanation		1 point
<i>Example: Some of the sheets of paper may be thinner than expected, so the actual capacitance for the circuit is larger than the measured capacitance.</i>		



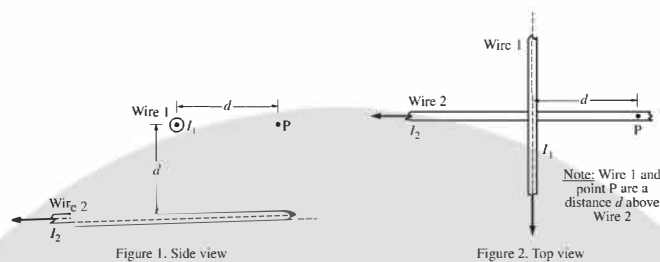


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Question 3

15 points total

Distribution
of points



The figures above represent different views of two long, straight, horizontal wires, 1 and 2, carrying currents $I_1 = I$ and $I_2 = 2I$, respectively, in the directions shown. The wires are held in place. In Figure 1, the current in wire 1 is directed out of the page, and wire 1 is a distance d above wire 2. Point P is a horizontal distance d from wire 1 and a distance d directly above wire 2. Express your answers to parts (a) and (b) in terms of I , d , and physical constants, as appropriate.

(a) 2 points

Use Ampere's law to derive an expression for the magnitude of the magnetic field at point P due to wire 1.

For attempting to use Ampere's law to calculate the magnetic field at point P	1 point
$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I$	
$B(2\pi d) = \mu_0 I$	
For a correct answer	1 point
$B = \frac{\mu_0 I}{2\pi d}$	

(b) 2 points

Derive an expression for the magnitude of the net magnetic field at point P.

For indicating $B_2 = 2B_1$	1 point
$B_1 = \frac{\mu_0 I}{2\pi d}$ & $B_2 = \frac{\mu_0 (2I)}{2\pi d} = \frac{\mu_0 I}{\pi d}$	
For an indication that the magnitude of the magnetic field is the vector sum of B_1 and B_2	1 point
$B_{net} = \sqrt{B_1^2 + B_2^2} = \sqrt{B^2 + B^2} = \sqrt{\left(\frac{\mu_0 I}{2\pi d}\right)^2 + \left(\frac{\mu_0 (2I)}{2\pi d}\right)^2} = \frac{\sqrt{5}\mu_0 I}{2\pi d}$	



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Question 3 (continued)

**Distribution
of points**

(c) 2 points

Calculate the numerical value of the angle to the horizontal for the direction of the net magnetic field at point P.

For correctly relating the angle to the individual magnetic fields	1 point
$\theta = \tan^{-1}\left(\frac{B_1}{B_2}\right)$	
For correctly substituting B_1 and B_2 into the equation	1 point
$\theta = \tan^{-1}\left(\frac{B_1}{B_2}\right) = \tan^{-1}\left(\frac{\left(\frac{\mu_0 I}{2\pi d}\right)}{\left(\frac{\mu_0 (2I)}{2\pi d}\right)}\right) = \tan^{-1}\left(\frac{1}{2}\right) = 26.6^\circ$	

(d) 2 points

Wire 1 is now released. Which of the following **best** describes the initial motion of wire 1 due to the magnetic field of wire 2? Assume gravitational effects are negligible.

- ☐ Wire 1 will not move.
☐ Wire 1 will move upward as viewed in Figure 1.
☐ Wire 1 will move downward as viewed in Figure 1.
☐ Wire 1 will rotate clockwise as viewed in Figure 2.
☐ Wire 1 will rotate counterclockwise as viewed in Figure 2.

Justify your answer.

For stating there is no translational motion since the magnetic forces on the wire cancel	1 point
For stating there is a net torque which causes rotation	1 point
<i>Example: In Figure 2, the top portion of wire 1 will be in a magnetic field into the page from wire 2 and, thus, will experience a force to the right. The bottom portion of wire 1 will be in a magnetic field out of the page from wire 2 and, thus, will experience a force to the left. So the net force will be zero, but there will be a net clockwise torque, so the wire will rotate clockwise.</i>	

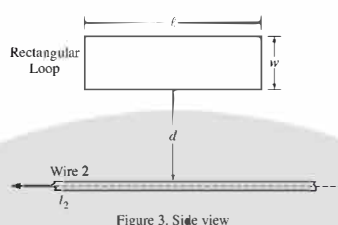




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Question 3 (continued)

**Distribution
of points**



Wire 1 is now replaced by a conducting rectangular loop of length ℓ , width w , and resistance R . The loop is placed a distance d from wire 2, as shown. The loop, wire, and distance d are all in the plane of the page. The long side of the loop is parallel to the wire. The current I_2 for wire 2 is decreasing linearly as a function of time t according to the equation $I_2 = 2I_0(1 - kt)$, where k is a positive constant with units of s^{-1} .

(e) 1 point

Of the following, select the integration that will give an expression for the flux Φ as a function of time t .

☐ $\Phi = \int_{r=d}^{r=d+w} \frac{\mu_0(2I_0)(1-kt)\ell w}{2\pi} dr$

☐ $\Phi = \int_{r=d}^{r=w} \frac{\mu_0(2I_0)(1-kt)\ell w}{2\pi} dr$

☒ $\Phi = \int_{r=d}^{r=d+w} \frac{\mu_0(2I_0)(1-kt)}{2\pi r} \ell dr$

☐ $\Phi = \int_{r=d}^{r=w} \frac{\mu_0(2I_0)(1-kt)}{2\pi r} \ell dr$

For selecting $\Phi = \int_{r=d}^{r=d+w} \frac{\mu_0(2I_0)(1-kt)}{2\pi r} \ell dr$	1 point
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Question 3 (continued)

**Distribution
of points**

(f) 3 points

Given that the flux through the rectangular loop as a function of time t is given by the equation

$\Phi = \frac{\mu_0 I_0 \ell (1 - kt)}{\pi} \ln\left(\frac{d + w}{d}\right)$, derive an expression for the magnitude of the current, if any, induced in the loop. Express your answers in terms of I_0 , d , r , R , w , k , ℓ , and physical constants, as appropriate.

For attempting to take the time derivative of the magnetic flux to calculate the emf	1 point
$\mathcal{E} = \left -\frac{d\Phi}{dt} \right = \frac{\mu_0 I_0 \ell}{\pi} \ln\left(\frac{d + w}{d}\right) \left \frac{d}{dt} [(1 - kt)] \right $	
$\mathcal{E} = \frac{\mu_0 I_0 k \ell}{\pi} \ln\left(\frac{d + w}{d}\right)$	
For dividing the emf by the resistance to calculate the current	1 point
$I = \frac{\mathcal{E}}{R} = \frac{\mu_0 I_0 k \ell}{\pi R} \ln\left(\frac{d + w}{d}\right)$	
For a correct answer	1 point
$I = \frac{\mu_0 I_0 k \ell}{\pi R} \ln\left(\frac{d + w}{d}\right)$	

(g) 3 points

What is the direction of the current, if any, induced in the loop as seen in Figure 3?

___ Clockwise ___ Counterclockwise

___ Undefined because there is no current induced in the loop

Justify your answer.

For selecting “Clockwise” with an attempt at a relevant justification	1 point
For indicating that the flux inside the loop will decrease	1 point
For using Lenz’s law to relate the decrease in the flux to the clockwise current	1 point
<i>Example: Because the current in the wire is decreasing, the flux in the loop will decrease. According to Lenz’s law, the induced current should create a magnetic field to oppose this decrease. Thus the induced magnetic field must be into the page, and the current in the loop must be clockwise.</i>	



